Independent adversarial referee report on `paper/clc.tex` (2026-10-29)

Produced by a reviewer agent given only the LaTeX source and instructed to re-derive every proof, test every claim numerically, and look for gaps. It did not read any other repository file. Each finding is followed by its resolution.

No FATAL findings. Theorems 2.1, 2.2, 2.3, 3.1(a),(b), Corollary 3.2, Lemma 3.4, Theorems 5.1–5.3 and the Corollary after 5.3 were found correct as written; Theorem 3.1(c) is true but its proof needed repair.

1. **ERROR — Theorem 3.1(c), Π > 1 branch.** "$(1+p_2^{-2})^3 \le 1+3.0001p_2^{-2}$" is false for every $p_2 \le 173$ (effective constant 3.00219 at $p_2 = 37$), and even granting it, $3.0001(8p_1^3-1) < 24p_1^3+3$ fails for $p_1 \ge 23$. Suggested fixes: use $(1+u)^3 \le 1+3u/(1-u)$, or prove $\Pi < 1$ rigorously. *Resolution:* the branch is now proved via "only the d-factor can exceed 1" ($e > d \ge 1$ and $s > 2p_1 \Rightarrow e, s \ge 2$), giving $\Pi \le 1+1/r^2$ and $r^2 \le 8p_1^3-1 < 24p_1^3$. Same error was found independently by the fact-check.
2. **GAP — Remark after 3.1 (Π < 1 always).** With $e = 2$ the stated inequality can fail (example p_2=1009, q=1021, r=1033); needs $e \ge 3$, which holds because $e = 2$ forces $Pr = q^2$. *Resolution:* the remark was removed; the proof no longer relies on $\Pi < 1$.
3. **ERROR (minor) — Lemma 3.3 "iff".** Solvability also needs gcd(p_1, L_{p_2}) = 1; counterexamples (13,157), (97,193). *Resolution:* statement now "iff $p_1 \nmid p_2^2-1$ and $p_1 \equiv p_2 \pmod g$"; the algorithm paragraph says such pairs are skipped (they have no completion). (Found independently in the author's own third pass before this report arrived.)
4. **GAP/TYPO — "$p_2 < \sqrt{24}\, p_1$^{3/2}".** Part (c) gives $p_2^2 \le 24p_1^3+1$. *Resolution:* now "$p_2^2 \le 24p_1^3+1 < 24.01p_1^3 = (4.9p_1$^{3/2})²".
5. **GAP (trivial) — Prop 2.3.** $s < 2\sqrt{2}p_2^2/(p_2^2-1) < 3$ needs $p_2 \ge 5$. *Resolution:* stated.
6. **GAP — Algorithm paragraph.** Range of the j-walk and the hand-over threshold unstated; $m > 2\lceil\sqrt{(2p_1)}\rceil$ unstated; strict vs weak window form. *Resolution:* paragraph rewritten with S, R_j, M*, explicit j-range and the exact m-window, the admissibility definition (weak form, as implemented), and the note that the two walks give identical survivor lists on six instances.
7. **GAP — Validation / proof of Theorem 3.5.** (a) Relaxed test cannot exercise the Fermat step (13685 is not Carmichael): say so. (b) Inconsistency "every survivor re-verified in Python" vs "survivors with $P > 2^{40}$": the $P \le 2^{40}$ survivors rest on the C 128-bit routine; re-verify all in Python. (c) "goes through with 2 replaced by κ" not literally true for part (c). *Resolution:* (a) stated; (b) text now says exactly what was done, the C routine is checked against Python in the test suite (`test_c_fermat_routine_matches_python`), and both campaigns are being rerun with `--dumpw` so that all 1,904 + 1,185 survivors are re-verified in Python (`clc/k4/rerun_dump.sh`, result in `RESULT_dump_reverify.txt`); (c) reworded to the lemmas actually used.
8. **STYLE/accuracy — Howe's definition.** Howe's criterion for order 2 allows $n \equiv 1$ or $n \equiv p \pmod{p^2-1}$; the paper's "iff $n \equiv p$" describes a subclass. *Resolution:* reworded; the CLC conclusion is

unchanged ($n \equiv 1$ is impossible for CLC n). Checking the exact wording against Howe's paper is on the pre-submission checklist.

9. **STYLE — §5.2** attribution of the side-split fact to Lemma 2.2. *Resolution:* now "immediate".
10. **STYLE — "x ≥ 2"** unnecessary. *Resolution:* "$x > 0$".
11. **STYLE — Theorem 5.2** even-m shortcut (optional). Not adopted.
12. **Heuristic.** Recomputed 6.3×10^{-7} / 96.5% (paper had 7.5×10^{-7} / 97%). *Resolution:* fixed, with the reproducing script `agrawal/k4_heuristic.py`.

What the referee verified numerically

Lemma 2.1 by brute force ($p < 120$, $k < 3000$); Prop 2.3 over all prime triples ≤ 3000 (max $t = 1.994$, max $s = 2.823$, no CLC triple); the product identity symbolically and the failure of the 3.0001 estimate exactly; the mod-24 step; impossibility of $p_2^2 = 24p_1^3+1$ for $p_1 \le 10^5$; Lemma 3.3 counterexamples by full-period brute force; Lemma 3.4's identity symbolically and all its inequalities on 62 relaxed ($\kappa = 24$, 120) two-prime tuples (true j not floor/ceil in 56 of 62); the relaxed four-prime problem (only 13685 below the search range; $d,e,s = 27, 67, 977$; $\mathrm{des}\cdot\Pi = 24^3\cdot5^3$); Theorems 5.1–5.3 symbolically for $m = 2..6$ and by brute force; the factorisations and Lucas conditions of 15500309690554865758009 and 26921089; Table 2's sum.